

Building capacity in the academic general practice research workforce

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Introduction

General practice provides most of primary care in the UK and is the cornerstone of the NHS, with well over a million GP consultations occurring each working day.¹ Lord Darzi's recent review of the NHS in England recommended three key policy shifts: from hospital to community care; analogue to digital; and treatment to prevention.² Globally, healthcare systems are recognising the importance of a strong general practice foundation to provide efficient and cost-effective health care to populations.³

High-quality research in general practice is vital to provide evidence-informed care to patients in the community, and the UK has led the world in developing academic general practice as a discipline. However, capacity within the academic GP workforce has been a longstanding concern, and two recent publications have highlighted the now critical situation. The Academy of Medical Sciences (AoMS) report *Future-Proofing UK Health Research*⁴ found that, while academic training posts in hospital specialties had fallen from 5% to 3%, academic GPs remained vastly under-represented at just 0.05% of the GP workforce, and the Medical Research Council (MRC) report *Clinical Researchers in the United Kingdom: Reversing the Decline to Improve Population Health and Promote Economic Growth*⁵ highlighted the diminishing number of academic GPs.

The paradox is that just as the NHS, and health systems internationally, aspires to a frame shift away from hospitals, academic general practice finds itself in a precarious position because of a declining research workforce. To address this crisis, the Medical Schools Council hosted a workshop with academic GPs at different career stages and Deans of medical schools from around the UK. The presentations and conversations from this event formed the basis of this article. Here we aim to summarise the achievements, history, and global context of academic primary care and the challenges facing academic GPs, then suggest priority areas as a starting point to spark conversations with policymakers, research funders, and the wider clinical GP workforce.

Academic primary care

For the UK, most academic primary care research is centred around academic GPs and their teams working in host departments in universities. Academic GPs need both clinical expertise and research skills to identify important questions that can be answered through studies grounded in primary care, so their job role usually spans both a university department and

NHS general practice. Primary care research led by academic GPs has directly informed guidelines, influenced policy, and improved patient outcomes both in the UK and globally. For example, as generalists, academic GPs recognised the presence of multiple long-term conditions in their patients requiring a holistic person-centred approach to management, and landmark studies using large GP databases were able to describe this at the population level.⁶ Routinely collected GP data provide real-world evidence to explore epidemiology of disease, risk prediction, drug surveillance, and, more recently, data-enabled trials. During the COVID-19 pandemic, large-scale platform trials led by academic GPs allowed rapid evaluation of treatments in community settings.^{7,8} However, despite these advances, clinical guidelines used by GPs are often based on research findings from specialist care, which can limit their applicability, further highlighting the need for more primary care-based research.⁹ To secure the potential to continue to grow policy and clinically relevant research in the future, we need a strong academic GP research workforce.

History and global context

At the beginning of the NHS in 1948, the University of Edinburgh took over the Edinburgh Dispensary Practice, where Sir James Mackenzie (1853–1925) was an early pioneer of primary care research, and this became the first independent Department of General Practice in the world.¹⁰ In 1963, Richard Scott was appointed by the University of Edinburgh as the first Chair of General Practice in the UK.¹¹ There was a significant expansion in the early 1990s thanks to NHS Research and Development and medical Service Increment For Teaching (SIFT) funding for general practices. Then the creation of the National Institute for Health and Care Research (NIHR) School for Primary Care Research (SPCR) in 2006 provided core funding for leading academic departments across England. NIHR funding, and similar schemes in the devolved nations, remain a major income source for university departments of academic general practice.¹²

The UK is widely recognised as the global leader in the field of academic general practice. Ian McWhinney, who moved from the UK to take up the first Chair of Family Medicine in North America, established the Department of Family Medicine at the University of Western Ontario in Canada in 1969. This tradition of academic migration has continued, exacerbating the workforce problem in the UK while enriching academic primary care internationally.¹³ As a universal healthcare system, the NHS provides a unique environment for research, serving

as a model for studying population health and service delivery in which the GP has a central role. UK primary care datasets have enabled large-scale studies that other nations could not easily replicate and the National Institute for Health and Care Excellence produces evidence-based guidelines that have an international impact on clinical practice.^{6,14}

Challenges in academic general practice

Despite this growth of general practice research, its world-leading reputation, and the renewed focus on prevention and earlier diagnosis and treatment outside of hospitals, academic general practice is in a fragile state. The longstanding challenges facing the academic GP research workforce have been previously described by the Society for Academic Primary Care,¹⁵ North American Primary Care Research Group,¹⁶ and leading scholars in the field, as well as in the recent AoMS⁴ and MRC⁵ reports.

A qualitative analysis of interviews with current GPs, GP registrars, and academic GPs found that GPs encountered barriers to both entering and remaining in the academic research workforce.¹⁷ The lack of visible and secure clinical academic career pathways and the financial uncertainty associated with short-term contracts in the early- and mid-career stages reduced the attractiveness of academia as a career option. The absence of pay parity between academic GPs and hospital consultants provided a further disincentive to pursuing a clinical academic career. Academic GPs may not reach the seniority required (typically senior lecturer or equivalent) to be paid on the consultant pay scale for many years. The discipline of academic general practice is also under-recognised compared with specialisation in competitions of esteem such as Clinical Impact Awards and, unlike consultant colleagues, there is no income protection for senior academic GPs who lose a national Clinical Impact Award. Academic GPs are also under-represented in Fellowship of the Academy of Medical Sciences and the Royal Society, and in the UK National Honours.

There has also been a loss of academic general practice as a visible department, and discipline, in many universities. Ironically, the interest that many academic GPs have in professional education and the high quality of teaching has resulted in many moving to medical education departments in universities and thus denuding academic research-based departments of primary care, a split that isn't commonly seen in other specialties.⁴ Moreover, despite the strong foundations on which the discipline of academic general practice is built, there is a lack of culture and precedent in pursuing a research career within general practice.

The lack of funding, infrastructure, and research culture makes academic GPs rare, and often different from their peers, needing unique support.¹⁸ A survey of GP registrars and trainers found that registrars had low intentionality to pursue research, and most did not engage in research activity during training.¹⁹ Trainers reported both not being engaged or interested in research and felt unprepared to mentor GP registrars in research. The contractual status of GP clinical academics is also variable across the UK. Clinical academic appraisals can be challenging as a result and although the Follett principles,

standards intended to support clinical academics, give guidance on joined-up job planning and appraisal across universities and the NHS, in practice this may not take place.²⁰ Moreover, the contribution of general practice research is under-recognised both within the general practice community and more widely.

Solutions for building capacity in the academic general practice workforce

A strong and sustainable academic general practice research workforce is needed to deliver evidence-informed care in the NHS and allow the UK to continue to lead on the global stage. To address the challenges summarised above and spark a wider conversation to build academic GP research capacity, we identified five key priority areas that require action.

Structured academic GP career pathways

The lack of clear academic general practice research career pathways creates uncertainty for potential and existing academic GPs, and the financial insecurity can lead to a loss of talent from the academic general practice pipeline. Funders and academic institutions need to work together to provide well-described routes for career progression, without the jeopardy of gaps in funding — particularly at the early- and mid-career stages — to recruit and retain young talent. Academic clinical fellow and clinical lecturer posts within general practice allow additional protected time for research and should be prioritised. There is also a need for pay parity for academic GPs with other clinical academics. Senior academic GPs should be paid according to the medical academic pay scales agreed by the British Medical Association (BMA),²¹ with equitable access to Clinical Impact Awards.

Academic general practice infrastructure

Universities need to recognise academic general practice as a discipline. There are many universities with small numbers of academic GPs often working on specialty topics and very isolated from other academic GPs. An infrastructure that allowed those small groupings to come together to make a sum greater than the parts might be a practical way of helping. For example, this could support collaborations and consortia of GPs from different institutions and areas bidding for research grants. Larger departments, such as members of England's NIHR SPCR and big academic primary care centres in the devolved nations, could also potentially support smaller neighbouring departments to strengthen and grow to ensure general practice research thrives across the UK and beyond.

A research culture within general practice

A culture change within general practice is necessary to make research, and the role of an academic GP, the norm. Medical students, registrars, and new GPs need to have experience of general practice research in both their formal teaching and experiential learning in GP practices. GP trainers also need support in mentoring research-active registrars.¹⁹ To deliver research in general practice, engagement is needed at all stages, from practices as patient identification centres to delivering an intervention and collecting outcomes, and arguably research should be within the remit of all GPs and their teams.

The WiseGP initiative in the UK aims to promote, advance, and sustain the distinct knowledge work of primary health care.²² The website and newsletter (<https://www.wisegp.co.uk>) raise awareness of the unique contribution of the generalist and provide practical wisdom for implementation of evidence-informed primary care tailored to patients, families, and communities. This work helps to bridge the gap between evidence generation and the daily application within general practice.

General practice has the shortest training of all specialties (with just 3 years as a specialist registrar) and this presents a challenge for finding time to do research. If GP training were extended, as proposed by the Royal College of General Practitioners (RCGP),²³ GP registrars would have more time to undertake research, present their findings at conferences, and publish articles. The work of academic GPs also needs to be visible and supported within the clinical team to increase and sustain research capacity within academic general practice.

Practical support

A career as a clinical academic can be demanding, with an academic GP sometimes having to juggle two competing roles.¹⁸ There is a need to recognise this and nurture those taking the route less travelled. Support can be in a variety of forms. Proactively identifying mentors, particularly for early- and mid-career researchers, could help ensure that clinical academics are supported through key career transition points.^{4,21} Peer-support through formal or informal action learning sets for those at similar career stages could also provide another source of wisdom and support. NHS England currently provides access to 'New to GP Fellowship' and 'Supporting Mentors' for new GPs, which could be adapted for academic GPs to improve recruitment and expanded to Northern Ireland, Scotland, and Wales. Practical help through clinical academic appraisals, including application of the Follett principles²⁰ to ensure a coherent job plan across both clinical and academic roles, should be supported by academic institutions, local NHS organisations, and NHS appraisal teams. The RCGP recently published a report on retention that summarises the workforce crisis in general practice and sets out required actions including calling for a national GP retention strategy; however, academic GPs were not mentioned.²⁴ Professional societies such as the RCGP and BMA need to champion the needs of the academic general practice research community and work with government departments, universities, and funders to ensure that academic GPs are included in plans for GP retention.

Contribution of academic general practice

General practice research can tackle key priorities for health systems globally in an efficient, pragmatic, and implementable way to allow evidence-informed care for patients. There is therefore a vital need to champion the discipline — wherever and whenever possible. The contribution of general practice research should be recognised by funders, academic institutions, peer-reviewed journals, and professional societies to improve the esteem in which the discipline is held. The *British Medical Journal* recently launched a 'Commission on the Future of Academic Medicine' and academic general practice should provide an important contribution.

Conclusion

Given the increasing importance of general practice within health systems globally, there is a need to encourage medical graduates to consider a career in academic general practice research and to support academic GPs at all stages of the clinical academic career pathway. A rapid and sustained expansion of the academic GP workforce is required to deliver the desired shift to evidence-informed care in the community. We hope this article encourages action by policymakers and research funders, and sparks discussion within the wider GP community, to ensure that the academic general practice research workforce can not only survive but also thrive in the future.

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